CANE WATER AUDIT

Discharge and Storage

Guidance for recording wastewater discharge, storage changes, spent wash and Zero Liquid Discharge (ZLD)

1. Purpose of this section

The Discharge and Storage section records where water and wastewater leave the plant, how much liquid is added to or removed from storage, how much spent wash is generated, and whether the distillery has a Zero Liquid Discharge (ZLD) system in place. Use measured daily data and a clearly defined reporting period. These entries describe different flows and should not automatically be added together.

Units: 1 m^3 = 1 kL = 1,000 litres. Thus 400,000 L/day = 400 m^3/day and 652,000 L/day = 652 m^3/day. The supplied values below are treated as the values entered in the audit form, not as verified or recommended operating figures.

2. Explanation of the four fields

A. Treated effluent discharged — 400,000 L/day

Meaning: The daily volume of wastewater that has undergone the plant's effluent-treatment process and is then discharged out of the defined site boundary or to the approved receiving point. The exact boundary matters: treated effluent sent to reuse within the plant is not the same as effluent discharged outside the site.

How to measure or record: Use a calibrated flow meter at the final discharge point, downstream of treatment and before the receiving drain, outlet or other approved discharge point. Record the start and end meter readings for the same 24-hour period. If more than one outlet exists, measure each outlet and sum the volumes. Record the discharge destination, sampling results and any abnormal events according to the plant's monitoring procedure.

Audit checks and improvement opportunities: Compare the discharged volume with the volume entering the effluent treatment plant (ETP), the volume reused internally and changes in effluent storage. Inspect for bypasses, leakage, stormwater entering the wastewater system and unmetered outlets. Review treatment performance and reuse options only where the water quality and applicable permissions allow.

Example: 400,000 L/day = 400 m^3/day. If the meter rises from 12,400 m^3 to 12,800 m^3 over a day, the recorded discharge is 400 m^3/day.

Important caution: The figure alone does not establish whether the discharge is permitted or compliant. Compliance depends on the actual discharge route, applicable consent conditions and measured effluent quality.

B. Change in storage (+ filling) — 0 L/day

Meaning: The net change in liquid volume held in relevant tanks, lagoons, reservoirs or other storage during the reporting day. A positive value means storage increased (net filling); a negative value means storage decreased (net drawdown); zero means there was no net change. This is a change in inventory, not necessarily a water discharge or consumption flow.

How to measure or record: Identify which storage units are included and use tank-level instruments, calibrated level-to-volume charts or reliable level readings at the same time each day. Calculate: end-of-day stored volume minus start-of-day stored volume. If several tanks are included, calculate each change and sum them. Use a consistent sign convention: positive for an increase and negative for a decrease.

Audit checks and improvement opportunities: Check storage levels against inflows, outflows, evaporation and transfers. Record transfers between tanks carefully so the same liquid is not counted as a net addition to site storage. A zero net change can occur even when substantial filling and withdrawal take place during the day.

Example: If storage begins at 1,200 m³ and ends at 1,200 m³, the net change is 0 m³/day = 0 L/day. This does not prove that there was no movement into or out of the tanks during the day.

Important caution: State whether this field covers raw water, treated water, wastewater, spent wash or all selected liquid-storage systems. Do not mix a change in storage with a daily flow total without using a complete water balance.

C. Spent wash generated — 652,000 L/day

Meaning: The quantity of spent wash produced by the distillery process during the reporting day. In many molasses-based distilleries, spent wash is the high-strength wastewater remaining after distillation; the exact stream and quantity depend on the process configuration and feedstock. Record the stream defined by the plant's audit boundary.

How to measure or record: Measure the spent-wash flow at a defined generation point using a suitable flow meter, or determine volume from calibrated collection tanks and batch records. Record the process type, operating hours, feedstock throughput and production for context. If the plant has more than one relevant stream, document which streams are included and avoid mixing spent wash with other wastewater unless the audit definition explicitly requires it.

Audit checks and improvement opportunities: Track spent wash generation against production and look for changes that may indicate process instability, dilution, leakage or measurement errors. Verify the capacity and operating status of the designated treatment, concentration, biomethanation, evaporation, incineration or other approved handling system, as applicable to the actual plant. Confirm that storage and transfer arrangements prevent uncontrolled releases.

Example: 652,000 L/day = 652 m³/day. This is a unit conversion of the entered value, not a typical or recommended generation rate.

Important caution: Spent wash is not equivalent to ordinary treated effluent. Its strength and required handling can be substantially different. Do not assume it can be discharged or reused without appropriate treatment and authorization.

D. Zero Liquid Discharge (ZLD) in place

Meaning: A yes/no status field indicating whether the plant has a functioning Zero Liquid Discharge system for the wastewater streams covered by the defined plant boundary. In practical terms, ZLD aims to prevent routine liquid effluent discharge outside the defined system boundary through treatment, recovery and management of residuals. The exact meaning must be consistent with applicable approvals and the plant's documented ZLD design.

How to measure or record: Select 'Yes' only after confirming the installed system's actual scope and operating status. Record the streams covered, system configuration, operating hours, recovered-water volume, final liquid discharge (if any), concentrate/sludge/salt management, bypasses and downtime. Keep supporting records such as flow-meter readings, operating logs, treatment reports and relevant regulatory documents.

Audit checks and improvement opportunities: Verify whether all relevant wastewater streams are included, including spent wash and other process effluents as applicable. Check whether any untreated or treated liquid is discharged off-site, bypassed, tankered away or sent to a separate facility. Review how reject streams and concentrated residues are managed and whether recovery systems operate as designed.

Example: The field value 'Zero liquid discharge in place' should be interpreted as a system-status claim, not as proof by itself that no liquid left the site on every day. Validate the claim against actual flows, operating records and the plant's approved boundary.

Important caution: If the audit form is a simple checkbox, retain an internal note explaining the scope of ZLD and any exceptions. A plant may have ZLD for one stream while other streams are managed separately; do not label the whole site ZLD without checking the defined scope.

3. Entered values — conversion and interpretation

Audit field	Entered value	Equivalent	Interpretation
Treated effluent discharged	400,000 L/day	400 m³/day	Daily measured liquid leaving the defined discharge point
Change in storage (+ filling)	0 L/day	0 m³/day	No net change in included

			storage volume
Spent wash generated	652,000 L/day	652 m³/day	Daily quantity of the defined spent-wash stream
Zero liquid discharge in place	Status field	Yes/No or configured option	Confirm actual system scope and operating evidence

4. Suggested daily record sheet

Parameter	Start / initial data	End / final data	Daily result	Evidence / remarks
Treated effluent discharged	Meter reading	Meter reading	End − start (L/day)	Outlet, destination, sample/log reference
Change in storage	Storage volume	Storage volume	End − start (+/− L/day)	Tank(s), level chart, transfers
Spent wash generated	Meter / tank reading	Meter / tank reading	Daily generated volume	Process, production, operating hours
ZLD in place	System scope/status	Operating status	Yes/No plus exceptions	Operating logs, recovery and residual records

5. Water-balance checks

- Keep generated spent wash separate from treated effluent discharged unless a documented treatment sequence proves they are the same stream at different measurement points.
- Do not add the spent-wash generation and treated-effluent discharge figures to obtain a site total without confirming that they are distinct, non-overlapping flows.
- Include changes in storage in the site water balance. A simple inventory convention is: net input − net output − net increase in stored liquid = unexplained difference, after accounting for evaporation, treatment losses and other defined terms.
- Investigate differences between wastewater generated, reused, stored, treated, discharged and removed as residuals. Document unmetered flows and uncertainty.
- ZLD status is not inferred from the other three numbers alone. Confirm system design, boundary, actual discharge records and management of concentrated residuals.

6. Example of how to use the values

The audit entries provided are: treated effluent discharged = 400 m³/day; net change in storage = 0 m³/day; and spent wash generated = 652 m³/day. These numbers should be reviewed together with the site's process flow diagram and wastewater-treatment arrangement. They do not, on their own, show that the site has a water-balance error or that ZLD is or is not achieved, because spent wash may be a separate stream, may undergo treatment, and may be measured at a different process boundary. Confirm the exact streams and boundaries before reconciling the totals.

Data quality note: Use plant measurements and the definitions in the audit tool. The document explains how to interpret and verify the fields; it does not certify the supplied values, regulatory compliance or ZLD performance.